

MATH 55 SECOND MIDTERM EXAM, PROF. SRIVASTAVA
APRIL 5, 2016, 3:40PM–5:00PM, F295 HAAS AUDITORIUM.

Name: _____

SID: _____

INSTRUCTIONS: Write all answers in the provided space. Please write carefully and clearly, in complete English sentences. This exam includes three pages of scratch paper at the end, which must be submitted, but will not be graded. Do not under any circumstances unstaple the exam. Write your name and SID on every page.

You are allowed to bring one letter-size single sided page of notes.

UC BERKELEY HONOR CODE: *As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.*

Sign here: _____

Question	Points
1	8
2	10
3	6
4	6
5	6
6	7
7	7
Total:	50

Do not turn over this page until your instructor tells you to do so.
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Name and SID: _____

1. Circle true (**T**) or false (**F**) for each of the following. There is no need to provide an explanation.

[Scratch space below]

Name and SID: _____

Name and SID: _____

[Scratch Paper 1]

Name and SID: _____

Name and SID: _____

3. (6 points) There are 5 distinguishable bins labeled $\{1, 2, 3, 4, 5\}$. How many ways are there of placing 100 indistinguishable balls into the bins, if each bin must have at least as many balls as its label? Be sure to explain your reasoning.

Name and SID: _____

4. (6 points) Show using a combinatorial proof that:

$$\binom{2n}{3} = \binom{n}{3} + \binom{n}{3} + \binom{n}{2}n + n\binom{n}{2}.$$

Name and SID: _____

[Scratch Paper 2]

Name and SID: _____

Name and SID: _____

6. (7 points) Prove that for every positive integer $n \geq 1$, the sum

$$S_n = 1 + 3 + \dots + (2n - 1)$$

is a perfect square.

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7. (7 points) Consider the function recursively defined by:

$$f(1) = 3 \quad f(2) = 2 \quad f(3) = 1 \quad f(k+1) = f(k) + f(k-1)f(k-2) \quad k \geq 3.$$

Prove that $f(n) \leq 2^{2^n}$ for all $n \geq 1$.

Name and SID: _____

[Scratch Paper 3]